

Impact on yard efficiency of a truck appointment system for a port terminal

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Abstract Port terminals consist of two interfaces for transferring cargo among transport modes: (1) the seaside or quayside interface and (2) the landside interface. At the seaside interface, cargo is loaded and unloaded from the vessels and stored temporarily at the yard. Landside operations consist of receiving and dispatching cargo from external trucks and rail. The increasing volumes of international trade are demanding more efficient cargo handling throughout the port logistic chain and coordination with the hinterland, hence attracting more attention from both practitioners and researchers on the landside interface of ports. Due to the high variability of truck arrivals with a significant concentration at peak hours, congestion at the access gates of ports and an unbalanced utilization of the resources occur. Truck

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appointment systems (TAS) have already been implemented in some ports as a coordination mechanism to reduce congestion at ports, balance demand and capacity, and reduce truck turnaround times. Based on the current situation faced by the Port of Arica, Chile, this paper aims to analyze potential configurations of a TAS and evaluate its impacts on yard operations, specifically in the reduction of container rehandles, as well as truck turnaround times. For this, a discrete-event simulation model and a heuristic procedure are proposed and experimentation is performed using historical data from the port terminal. Results indicate that implementing a TAS may significantly benefit yard operations in terms of reducing container rehandles as well as truck waiting times.

Keywords Port terminal operations · Truck appointment system · Discrete-event simulation · Landside coordination

1 Introduction

Port terminals consist of two interfaces for transferring cargo among transport modes: (1) the seaside or quayside interface and (2) the landside interface. At the seaside interface, cargo is loaded and unloaded to/from the vessels and stored temporarily at the yard. Landside operations include receiving and dispatching cargo from external trucks and rail. The increasing volumes of international trade are demanding more efficient cargo handling throughout the port logistic chain and coordination to the hinterland, thus attracting more attention from both practitioners and researchers on the landside interface of ports. Intermodal transport chains are more complex nowadays, with the introduction of bigger container ships and the new configurations of shipping services, with decreased frequency but bigger volumes of container transfer at the ports. Hence, not only is infrastructure needed to support operations at the quay, but also stacking capacity in the yard as well as appropriate infrastructure connecting to the hinterland. These trends impose important challenges to the ports in terms of a better coordination of the port logistic chain, implying a more active role, beyond the traditional role as an intermodal interface for the transfer of cargo (Rodrigue 2012).

High levels of congestion at the gate of the ports and in the road access are affecting inland costs of foreign trade, implying less productivity of trucks and rail. Several ports have faced important challenges in terms of congestion, with high waiting times and inefficient cargo handling operations. As a solution to mitigate congestion at the ports, some ports worldwide have implemented technology platforms, known as Truck Appointment Systems, to support the coordination of the inland flows. The first references are the ports of Los Angeles and Long Beach that during the early 2000s implemented different strategies to mitigate congestion (Giuliano and O'Brian 2007).

In the mid-2000s, congestion at the roads and gates of the port terminals in Australia motivated the ports to invest in technology and infrastructure to support the coordination of inland flows. Landside operations were significantly improved due to the implementation of a landside improvement strategy, where a Vehicle Booking System (VBS) was implemented for the two stevedores (Patrick and DP World) in the five main port systems in Australia. A booking system differs from an appointment system in that the truck carriers in the latter are assigned an appointment (mainly for import operations in practice) with the specific hour at which they should arrive to the container terminal. A booking system is more flexible in that the truck carriers can select a slot among available time zones in the port terminal. The VBS implemented in Australian ports was very effective in terms of smoothing the truck arrival pattern, minimizing peak periods and increasing the productivity of the operations

for both the truck carriers and stevedores. Additional efforts were conducted, including the implementation of a Task Force where the stakeholders converse and meet regularly to discuss and plan landside issues, infrastructure needs to support peak hours (although they are reduced with the VBS, they are not completely eliminated), truck marshalling areas, and notification systems for truck arrivals at the empty container parks.

In this work, we present a simulation model to estimate the potential impact of implementing an appointment system at a port terminal where congestion due to the lack of coordination of inland flows is generating long waiting and service times for the trucks, as well as inefficient cargo handling operations in the yard of the port terminal. In order to assess potential benefits, we develop a discrete-event simulation model to evaluate different strategies for the configuration of the appointment system, considering the particular case of the Port Terminal of Arica in Chile. In particular, we compare the current system where random arrivals occur with a truck appointment system analyzing the truck waiting time (average and maximum) and the number of container rehandles. Furthermore, we evaluate different TAS schedule policies combined with different ways to assign appointments to trucks. In addition, we perform a sensitivity analysis to study up to what extent the appointment system can keep the gains in the operations depending on the percentage of arrivals. We consider the case of an appointment system rather than a booking system as a first approach, but further research can extend the analysis to a booking system as in the Australian case.

The contributions of this paper are: (1) the analysis of potential benefits of an appointment system, considering as performance metrics the impact on yard operations as well as congestion indicators that previous work in the literature have not considered simultaneously (e.g., [Zhao and Goodchild 2010](#); [van Asperen et al. 2013](#)), (2) the definition and evaluation of two schedule policies for the appointment system, and (3) the definition and evaluation of policies for the assignment of containers to appointments. In order to evaluate the appointment system and the assignment policies, different scenarios based on real information provided by the port terminal of Arica are evaluated, considering only export operations as a first scope of the analysis. Our scope also differs from the scope of previous works on appointment systems which have been mainly focused on import operations. In terms of the methodology, considering export or import containers does not imply a significant difference. However, physical and documental procedures related to export and import containers significantly differ, as it will be further described herein.

This manuscript is structured as follows. Next we provide a brief literature review. Then, Sect. 3 presents background information on the current operations and characteristics of the Port of Arica. Section 4 describes the modeling approach along with details of the simulation model. Section 5 describes the experiments and analyzes the results. Section 6 provides managerial insights and discusses the challenges for implementing a TAS at the Port of Arica. Finally, Sect. 7 presents the concluding remarks and explores potential extensions for further research.

2 Literature review

We can find several contributions in the literature that address different decision problems arising at port terminals and the whole port logistic chain. As port terminals are divided into three main areas (waterside, yard and landside), there are several contributions that address different problems, either applying a hierarchical approach or considering an integrative approach. Ship planning decisions consider berth allocation, quay crane assignment and scheduling, and stowage planning (e.g. [Avriel et al. 1998](#); [Cheong et al. 2010](#); among

others). [Bierwirth and Meisel \(2010\)](#) present a survey of the contributions related to the berth allocation and quay crane scheduling problems and provide a classification scheme. Yard decisions consider the assignment of containers to positions in the yard and the planning and scheduling of yard equipment. At the gate, decisions consider capacity planning in terms of defining the number of trucks to receive per period, as well as demand forecasting and gate-lanes layout. More from a supply chain perspective, some decisions related to hinterland coordination and network design are also considered. A review of this area is provided by [Lam and Gu \(2013\)](#). For a detailed survey on the contributions of operations research approaches to container terminals, the reader is referred to [Steenken et al. \(2004\)](#) and [Stahlbock and Voß \(2008\)](#).

Truck appointment systems have been studied in the literature by several authors in order to estimate potential benefits obtained considering different case studies. Another research avenue has focused on the design of models to support gate capacity and truck scheduling for drayage operations under the operation of a truck appointment system. We present a review of the most relevant literature in this topic and provide some insights of the existing work and the contribution of our research.

[Giuliano and O'Brian \(2007\)](#) present a study to evaluate the potential impacts of the truck appointment system implemented at the Ports of Los Angeles and Long Beach in the United States, which are some of the first cases reported in the literature addressing these issues. They reported that not all the expected benefits were achieved by the truck carriers. [Huynh and Walton \(2008\)](#) examine the impact that limiting truck arrivals may have on yard crane utilization and truck turnaround times. They propose a methodology to determine the optimal level of truck arrivals per time window.

[Davies and Principal \(2009\)](#) evaluate the improvements obtained with the implementation of a truck appointment system at the Ports of Botany (Sydney, Australia), Vancouver (Canada), Southampton (UK), and Long Beach (US). They observed that the implementation of the system had benefits both for the ports and truck carriers, as congestion at the gate is minimized. [Zhao and Goodchild \(2010\)](#) analyze the impacts of having truck arrival information on the operations in the yard, specifically container rehandles. They propose two heuristic procedures to reduce rehandles at the yard. The first one involves determining the storage location of rehandled containers to avoid future rehandles, and the second procedure consists on dictating the container pickup sequence for trucks so that it matches the container stacking sequence as much as possible. The second procedure shares some similarities with a truck appointment system, but differs in that trucks do not book an appointment, and are just organized into a sequence once they arrive to the port terminal so that rehandles may be minimized.

The strategy described above is appropriate for ports with enough space to re-arrange the queue and sequence of trucks for the dispatching of containers, which is not the case in most of the port terminals in developing countries where space is very limited. Otherwise, congestion at the access to the port may significantly increase. [Zhao and Goodchild \(2010\)](#) provide a simulation model in which different scenarios based on the availability of truck arrival information are used to estimate the impact on the number of rehandles in the yard. Their results indicate that it is not needed to provide the exact sequence of truck arrivals to achieve important reductions of rehandles in the yard, and that aggregated information may significantly improve yard operations. Previous works do not consider any congestion measure or any performance indicator with respect to truck productivity as is considered herein, which is an important performance metric for this type of coordination systems.

[van Asperen et al. \(2013\)](#) also study the impact of truck arrival information on yard operations. They consider the strategy of truck announcements, which differs from a truck

appointment system or a pickup sequence dictation [as defined by [Zhao and Goodchild \(2010\)](#)] in that trucks announce their arrival to the port terminal with some hours of anticipation. This strategy is different, as the truck arrivals pattern is not changed (the same arrival time flexibility is maintained), with the only difference being that the port terminal is notified some time in advance. Although significant benefits can be achieved by providing information in advance to the port terminal, a notification system does not necessarily reduce congestion at the access and within the port. Furthermore, it requires high flexible capacity at the port terminal in order to be able to re-arrange containers in the yard with the information provided in real-time, as well as determining the sequence of trucks at the gate (that may be re-arrange). In addition, this system is only feasible if the port terminal has enough space to permit trucks to be re-arranged, which is not the case for most of the Latin American ports.

As in this manuscript, among others, we are concerned with the number of rehandles of containers that result when containers are retrieved from the blocks in the yard. We can note that this problem is related to a specific combinatorial optimization problem known as the blocks relocation problem [(see, e.g., the surveys in [Caserta et al. \(2011\)](#) as well as [Lehnfeld and Knust \(2014\)](#)]. Nevertheless, our main concern is not this somewhat stylized problem treated for just one bay rather than in the general context as it is needed here. This is also in line with the above references provided.

[Gracia et al. \(2016\)](#) propose a simulation model to design the configuration of a Booking System for the Port of San Antonio in Chile. The aim of this system is to provide a technological platform for coordinating the inland flows of trucks to the gate of the port terminal. In order to estimate potential impacts in terms of congestion, factors such as gate lane segmentation and booking levels were considered to estimate the impacts in terms of gate congestion (number of trucks at the gate and waiting times). Particularly, for the case of the Port of San Antonio, the gate has several lanes that could be assigned to different types of flows. It is also possible to consider a fast track lane for those trucks with appointments, considering that the research considered the evaluation of a gradual implementation of a TAS. For the research proposed in this work, the Port of Arica has only two lanes for the gate-in and gate-out of trucks and we are considering the situation in which the truck appointment system is fully implemented, which implies that no truck can arrive to the terminal without an appointment. In addition, the research presented herein extends previous work by analyzing the impact on yard operations and not only on the congestion of the gate.

Contributions in the literature related to supporting gate capacity planning of a TAS have proposed mathematical models that aim to support decisions related to the number of trucks to receive at each time period. Examples of this can be found in ([Chen and Yang 2010](#); [Chen et al. 2013](#); [Zehendner and Feillet 2014](#); among others). Collaborative schemes for determining the appointments among truck carriers and the port terminal have been proposed by [Phan and Kim \(2015; 2016\)](#) in which pricing mechanisms are employed rather than traditionally setting a capacity constraint on the number of trucks to be received. On the other hand, a collaborative truck appointment system to reduce empty truck trips and enhance horizontal collaboration among truck carriers is presented by [Schulte et al. \(2015\)](#).

Although truck appointment systems have been implemented in several ports worldwide, not all the expected benefits have been achieved. For instance, [Giuliano and O'Brian \(2007\)](#) reported that no evidence of significant reduction on truck waiting times was achieved when the system was implemented at the ports of Long Beach and Los Angeles. On the other hand, [Islam and Olsen \(2014\)](#) highlight the challenges faced by truck carriers when TAS were implemented at the ports in New Zealand. These challenges were related to reducing the flexibility of trucks and the possibility of sharing capacity among carriers. Hence, operating policies are factors that significantly impact the benefits that can be achieved when this type of

systems is implemented, as these are solution mechanisms that affect the operations of several stakeholders of the port. Thus, it is very important to consider the specific characteristics of each port. Accordingly, in this manuscript we consider a case study in an emerging country with very limited capacity and special conditions that make operations very complex for the managers of the port terminal. The provided analysis aims to study the potential benefits achieved in terms of the efficiency of yard operations as well as truck congestion so as to provide insights of the benefits expected and some recommendations in terms of container stacking procedures that may result in lower rehandles of containers at the yard. Furthermore, we provide a discussion on the challenges to overcome in order to implement a TAS such as the one described in this paper.

3 Port of Arica

The problem analyzed in this paper is inspired by the operations of the Port of Arica in Chile. The Port of Arica has only one terminal, known as TPA by its acronym in Spanish. The Port of Arica is positioned one of the 10 most important ports of Chile in terms of the volume of TEUs (Twenty Foot Equivalent Units) transferred, occupying the 6th position at a national level and the 43rd position at the region of Latin America and the Caribbean, according to the *Latin American and the Caribbean Container Port Throughput Ranking 2015* published by ECLAC/UN¹ (Economic Commission for Latin America and the Caribbean, United Nations) in 2015. This ranking is elaborated using public information submitted by ports, terminals and national port authorities to ECLAC in terms of the TEUs transferred over a specific time period.

TPA has important challenges in terms of the landside operations, which refer to the coordination of trucks that arrive to the port for picking up and delivering cargo. Due to the increasing volumes of cargo, the port terminal is facing significant congestion levels, where dwell times are very high compared to other ports in Chile and worldwide, due to the special conditions of the port, which transfers mainly (more than 70%) in-transit cargo from Bolivia. Chile and Bolivia maintain political agreements in which it is stated that the cargo that is transferred by the ports in the north of Chile (Arica, Iquique and Antofagasta) has free transit and free storage cost for 60 days for export cargo and 360 days for import cargo. In addition, customs and other authorization procedures for the cargo are not made electronically and nor with anticipation of the arrival of the ship, implying significant delays for the dispatching of import cargo. Import containers are more difficult to handle at the yard compared to export containers that are segregated according to the ship to be loaded, the port of destination and weight. In contrast, import containers are not segregated because the port managers do not have any criteria to apply due to the lack of coordination mechanisms with the consignees and cargo agents. Bolivian in-transit cargo presents high uncertainty and variability with respect to dwell times, which is enhanced due to the condition that Bolivian in-transit cargo has 360 days of free storage. Thus, the consignees are not under pressure to pick up the containers as it occurs in other ports.

We focus the analysis on the case of export operations, rather than imports as a first stage of the analysis. In terms of the methodology, there is no significant difference, but the operations of this type of cargo differ greatly. Export containers arrive randomly prior to the arrival of the ship, and specifically for the case of the Port of Arica, Bolivian transit cargo is allowed

¹ <http://www.cepal.org/cgi-bin/getProd.asp?xml=/perfil/noticias/noticias/4/54974/P54974.xml&xml=/perfil/tpl/p1f.xml&base=/perfil/tpl/top-bottom.xml>.

Tier 5	5					
Tier 4	4					
Tier 3	3	8				
Tier 2	2	7				
Tier 1	1	6				
	Stack 1	Stack 2	Stack 3	Stack 4	Stack 5	Stack 6

Fig. 1 Placement of containers in a bay according to the arrival sequence

a free-transit condition so they can arrive at the port even if they have not been assigned to a vessel. The sequence on which export containers arrive to the port has a direct impact on the ship turnaround time, which is one of the most important key performance indicators of the port terminal. On the other hand, import containers are requested once they are unloaded from the ship and temporarily stacked in the yard of the terminal. Due to the free storage condition of import cargo, it is very hard for port managers to determine with anticipation the moment at which containers will be requested by the cargo agent. Hence, port managers recommended us to focus on the analysis of export containers as a first step.

The export process begins when the truck carrier arrives at the gate. The gate personnel verifies the information of the container in the system and the truck is directed to the yard where the container will be stacked. Currently, there is no pre-determined schedule for the arrival of trucks and no information on the arrivals is available for the port terminal with anticipation, hence the time required for the service depends on the congestion level and the available resources at the port terminal.

The yard layout considers a set of blocks for import containers and another set of blocks for export containers. There is also an area referred to as the Cargo Freight Station (CFS) for stuffing and de-stuffing cargo. We do not include in the analysis the empty containers that arrive to the port terminal for stuffing operations, as they are handled in the CFS. Export and import areas consist of dedicated blocks and import and export containers are never located within the same block.

Each block of the yard consists of three coordinates which are referred to as the BaRoTi system (Bay, Row or Stack, Tier). Each export container should be assigned to a position in a block of the export area, indicating the bay, stack, and tier coordinates, which are registered in the tracking device by the yard operators. Only containers of the same size (20 or 40 ft) that will be loaded in the same ship are assigned to a single bay. Upon arrival of a container, the stack and tier are defined based on the arrival sequence. Thus, containers are placed starting at one extreme of the block at the ground level, then successive containers would be piled up in the same stack until the last tier is used. Then, the next container would be located in the next stack. Figure 1 shows how eight successive containers (1 arrives first, 8 arrives last) that would be loaded onto the same ship are placed.

A rehandle occurs when a container (target) needs to be retrieved from the yard in order to load it into the corresponding ship, and there are other containers interfering and blocking the target container. Therefore, these interfering containers need to be moved and relocated. Typically, these containers are located at the same bay and stack, but in a lower tier once the target container has been retrieved. We assume this for all relocated containers as this is a common practice of the yard managers.

In order to handle full export containers, TPA has 8 reach stacker vehicles (also referred as cranes in this paper) assigned to handling operations. This type of crane is able to reach a container from both ends of the bay (stacks 1 and 6 in Fig. 1). Furthermore, the reach stacker is able to handle a container located up to 5 positions from the ground level measured as

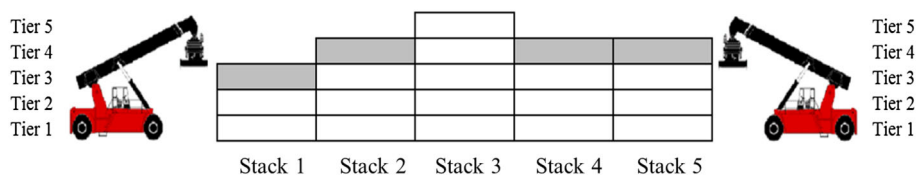


Fig. 2 Illustration of containers can be removed without rehandles (*gray*)

a rectilinear distance without incurring in rehandles. For instance, Fig. 2 shows in gray the containers that can be accessed without rehandles.

Based on the current situation as well as the statistics reported by the port terminal, truck turnaround times are on average 80 min for export containers, but up to 4 h for import containers due to documental procedures that are required to be fulfilled. In previous years, this could take up to 14 h. The reader should notice that, for the scope of this paper, we focus only on export operations as it has been pointed out before. On the other hand, the average number of rehandles per container are in the range of 1.5 and 3, depending on the conditions (if this is an import or export container, and in the case of import containers, it also depends on the dispatching procedure).

Due to the complexity of the operations at a port terminal and the uncertainty on the inland flows, there is variability on the truck arrivals. Given the variability of the operations at the port and potentially disruptive events that may affect the fulfillment of an appointment, we model arrivals with a probability-based deviation from the appointment time, which implies that some trucks may arrive earlier or later than the appointed time. Furthermore, the deviations from the appointment time are represented using several continuous probability distributions that allow the experimentation with different scenarios. Early or late arrivals are due to the variability of the operations at the different echelons of the port logistics chain and, for the case of the truck carriers, they are subject to delays in previous stages (such as the facility of the importer where the cargo is consolidated) or accidents and other delays on the road. No major disruptive events such as natural disaster or any other situations that interfere the operations at the port terminal are considered. For a detailed study of disruptive management for a TAS the reader is invited to consider the work presented by Li et al. (2016).

4 Modeling approach

In order to evaluate the impact of a TAS on the congestion at the gate and also on the operations at the yard, we propose a two-stage modeling approach. First, a discrete-event simulation model of the gate is developed to evaluate the impact of a TAS on the gate congestion. In addition to performance measures related with truck waiting, the simulation model produces, as another output, the arrival sequence of containers that will be placed in the yard. This sequence feeds a heuristic procedure to locate the containers in the yard and count the number of container rehandles based on the real loading sequence observed at TPA.

Figure 3 shows the modeling approach and the relation between models. Discrete-event simulation models allow one analyzing complex systems where variability plays an important role and evaluating multiple system design alternatives under realistic conditions. Therefore, this type of model is suitable for analyzing the port operations. Furthermore, the heuristic procedure devised for this paper is able to replicate the allocation and retrieval of containers at the yard, which makes it suitable for the scope of the paper.

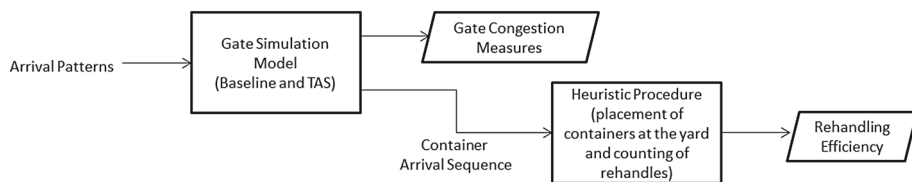


Fig. 3 Modeling approach to evaluate TAS impact on gate and yard operations

4.1 Discrete-event simulation model of the gate

The first-stage of the modeling approach compares two scenarios of the system under study: the current no-TAS situation (baseline) with a TAS that allows the port to smoothen the arrival rate over a day. The comparison of both scenarios is based on gate congestion measures, such as the mean and maximum truck waiting time, mean and maximum truck queue length and crane utilization.

The simulation model is based on real data of four days of operations at TPA. Each day, the TPA operates 16 h, from 7:00 to 23:00. The total number of trucks with a full export container that arrived during those four days is 658. Upon arrival, the truck can be served only if a crane is available. Based on TPA operations, we assume that there are 8 cranes assigned to locate full export containers in the yard. If all cranes are busy, the truck must wait in a queue based on a First Come First Served basis. The processing time for the crane includes the lapse from the removal of the container from the truck to the location of the container in the yard. In order to determine the probability distribution that better off represents the processing time, Chi-square and Kolmogorov–Smirnov (KS) goodness-of-fit tests were used over a sample of 579 data points with sample mean and sample standard deviation of 25.23 and 17.21 min, respectively. Based on this analysis, the best fit for the processing time is a gamma distribution shifted +1 min with parameters $\alpha = 2.12$ and $\beta = 1/12.04$, whose mean and standard deviation are 25.22 min and 17.07 min, respectively. The p values for the Chi-square and KS tests are 0.134 and 0.363, respectively, which implies that the hypothesis that the sample data comes from a gamma distribution with the indicated parameters cannot be rejected.

Figure 4 shows the histogram of the sample data and the probability density function of the fitted distribution for the crane processing time.

Since a TAS has not been implemented at TPA yet, the input data for the scenarios with a TAS considers the experience observed by the Port of Sydney, Australia where a TAS is used in its operations and reports are periodically published on its website.² In particular, the distribution of arrivals of trucks over a day with a TAS was estimated based on the operations of the Port of Sydney.

Hence, the operation hours of TPA are classified into three categories based on the traffic intensity: peak, shoulder, and off-peak. Two periods of peak hours have been identified: from 11:00 to 13:00 and from 16:00 to 18:00. Just before and after the peak hours, the shoulder periods take place: from 9:00 to 11:00, from 13:00 to 16:00 and from 18:00 to 20:00. Off-peak periods occur from 7:00 to 9:00 and from 20:00 to 23:00. Currently, 51% of the arrivals in a day occur during the peak hours, 43% occur during the shoulder periods and only 6% during the off-peak periods.

² <http://www.sydneyports.com.au/>.

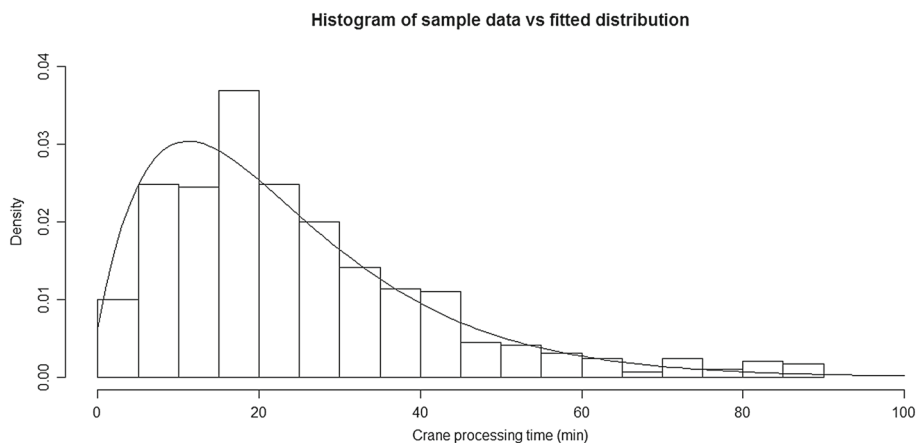


Fig. 4 Histogram of sample data vs fitted distribution of the crane processing time

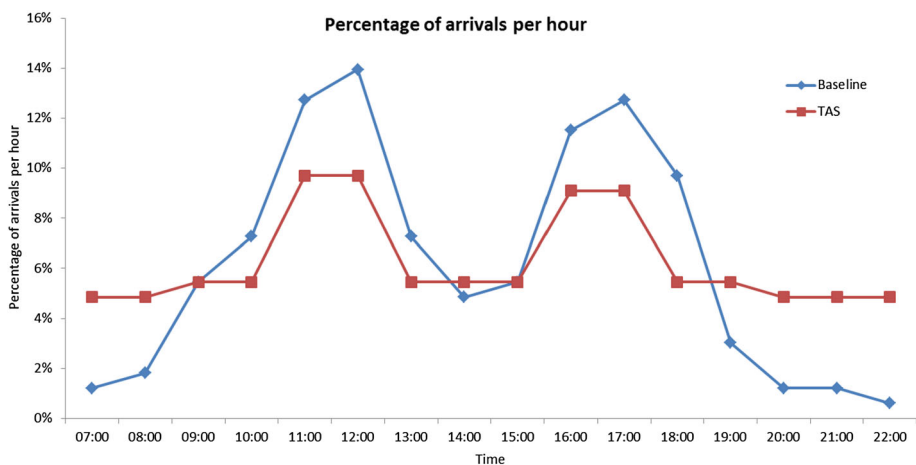


Fig. 5 Percentage per hour of daily arrivals of trucks

Based on the experience of the Port of Sydney, the simulated system with TAS redistributes the arrivals over a day such that 38% of the arrivals take place during the peak periods, 38% of the arrivals occur during the shoulder periods and 24% of the arrivals are scheduled for the off-peak periods. Figure 5 shows the mean percentage of arrivals per hour for the baseline and the TAS.

Furthermore, the simulation of the TAS considers two schedule policies:

- TAS/Day-Even, where the truck arrivals are evenly distributed across the four days of simulation.
- TAS/Day-Var, where the number of truck arrivals per day varies as observed in the real system (baseline), but smoothed across the hours of the day based on the percentages shown in Fig. 5.

Table 1 shows the number of arrivals per day in the simulation scenarios.

Table 1 Number of arrivals of trucks per day

Scenario	Day			
	1	2	3	4
Baseline	149	292	111	106
TAS/day-even	165	165	165	165
TAS/day-var	149	292	111	106

A non-stationary Poisson process is assumed for the arrival of trucks with mean equal to the observed arrivals for the baseline scenario and proportional according to the TAS for the scenarios with the appointment system.

The simulation of the TAS assumes that each truck is assigned an appointment time (e.g. 8:00), but if the truck arrives within the following hour of the appointment (e.g. between 8:00 and 9:00), then the arrival is considered to be on time. There are 16 appointment slots in a day, which correspond to each hour between 7:00 and 23:00. In the experimentation section, we describe the different scenarios simulated to test for deviations of truck arrivals from the appointment time.

4.2 Heuristic procedure for container location and rehandle counting

The arrival sequence obtained from the simulation model feeds a heuristic procedure that places the container in the yard and counts the number of rehandles based on the description shown in Sect. 3. Current practice at TPA places export containers in the yard based on segregations, which are groups of containers with similar characteristics (e.g., size and assigned ship). Therefore, in order to reduce rehandles, containers of the same segregation are ideally located near each other in the yard.

The heuristic procedure assigns the container to a block and bay dedicated to store containers of the same size and ship. Then, stack and tier are defined based on the arrival sequence as described in Sect. 3. Based on the layout of the yard at TPA, we assume that all bays have 6 stacks and 5 tiers.

The arrival sequence generated by the simulation model for the different scenarios allows the heuristic procedure to locate the containers at the yard. Then, considering the real loading sequence as another input, the heuristic procedure counts the number of rehandles incurred by the system when these containers are loaded on the ship. The resulting number of rehandles is based on the minimum number of movements required when evaluating the retrieval of the container from both sides of the bay.

5 Experiments and results

In this section we describe the experiments that were performed in order to assess the potential benefits of a TAS at TPA. The discrete-event simulation model for the gate operations was developed in Simio 8.132 and the heuristic procedure was coded in VBA for Excel. Computations were run using a 1.50 GHz Intel Core i5-3339V processor with 4MB of RAM.

5.1 Deviation from appointment time

As stated before, the discrete-event simulation model of the gate compares the performance of the current system (baseline) where truck arrivals occur randomly with a TAS where

appointment times are distributed according to the policies described in Sect. 4.1. In addition, two schedule policies for TAS are evaluated, TAS/Day-Even and TAS/Day-Var where the former assumes that the number of arrivals is evenly distributed across different days (165 arrivals/day) while the latter assumes that the number of arrivals per day varies according to the observed arrivals in the real system (baseline).

The simulation of the baseline scenario generates the arrivals at the exact time that occurred in the real system. However, for this scenario, the crane processing time is a random variable with probability distribution $1 + \text{Gamma}(2.12, 1/12.04)$ min.

For the experiments based on a TAS, we define several scenarios for the deviation of the truck arrival from the appointment time. The deviation from the appointment time is represented through a probability distribution function. All distributions selected for experimentation have a mean equal to 0.5 h. Therefore, the expected value of the arrival time is half way of the hour that is considered as on-time arrival. For example, for all the trucks whose appointment time is 9:00, the expected value of the arrival time is 9:30. If the arrival occurs between 9:00 and 10:00, then the truck is on-time. Table 2 shows the scenarios defined for the deviation from the appointment time.

In order to better understand the scenarios in Table 2, consider the following:

- The column *Distribution* shows the name and parameters of the tested distribution: Uniform (min, max), Normal (μ, σ), Gamma (α, β) and Exponential ($1/mean$).
- The units of the parameters of the uniform, normal and exponential distributions are hours.
- The gamma and exponential distributions are lower bounded. Therefore, for scenarios g, h, i and k , we use these distributions with a shift of -1 h. We limit the early arrivals to one hour because they are more easily to be controlled by the truck driver than the late arrivals.
- The mean for the uniform distribution is $(min + max)/2$ and for the gamma distribution is $(\alpha/\beta) + shift$.
- The standard deviation of the uniform distribution is $(max - min)^2/12$, for the gamma distribution it is $(\alpha/\beta^2)^{1/2}$ and for the exponential distribution it is equal to its mean.
- The coefficient of variation is the ratio of the standard deviation to the mean. The scenarios include distributions with a relatively low variability (scenarios b, e and g) and distributions with high variability (scenarios f, j and k).
- The last six columns show the probability that an arrival occurs on time (during the 1 h assigned slot), before the appointment time, after the appointment time (later than the 1 h assigned slot), before the appointment time but no earlier than 1 h before, between 1 and 2 h after the appointment time and more than 2 h after the appointment time (2 h later than the end of the 1 h assigned slot).
- The short name for the scenarios includes the first letter of the probability distribution, the on-time probability and the shift (if it applies).

Table 2 shows a variety of scenarios that include the following characteristics:

- Scenario a assumes that 100% of the arrivals occur on time (during the 1 h assigned slot).
- Scenarios b to j model an appointment compliance of 75%, 85% and 95%.
- Scenario k is an extreme case where appointment compliance is only 11.7%.
- Scenarios e, f and g assume that the probability of having an early arrival is 0%.
- Scenarios b, c, g, h, i and k allow trucks showing up early up to 1 h before the appointment time. Scenario d also enables early arrivals, but the probability of a truck showing up before 1 h the appointment time is only 0.0003.

Table 2 Scenarios designed to test for the deviation from the appointment time

Scenario	Distribution	Short name	Mean (h)	SD (h)	C. V. (SD/mean) (%)	Probabilities					
						On time	Early arrival	Late arrival	Early arrival (up to -1 h)	Late arrival (between 1 and 2 h)	Late arrival (more than 2 h)
<i>a</i>	Uniform(0, 1)	U(100%)	0.5	0.2887	57.74	1	0	0	0	0	0
<i>b</i>	Normal(0.5, 0.2550)	N(95%)	0.5	0.2550	51.00	0.95	0.0250	0.0250	0.0250	0.0250	0
<i>c</i>	Normal(0.5, 0.3473)	N(85%)	0.5	0.3473	69.46	0.85	0.0750	0.0750	0.0750	0.0750	0
<i>d</i>	Normal(0.5, 0.4346)	N(75%)	0.5	0.4346	86.92	0.75	0.1250	0.1250	0.1247	0.1247	0.0003
<i>e</i>	Gamma(3.56, 7.12)	G(95%)	0.5	0.2650	53.00	0.95	0	0.0500	0	0.0498	0.0002
<i>f</i>	Gamma(0.698, 1.396)	G(85%)	0.5	0.5985	119.6	0.85	0	0.1500	0	0.1181	0.0318
<i>g</i>	-1 + Gamma(34.02, 22.68)	-1 + G(95%)	0.5	0.2572	51.43	0.95	0.0155	0.0345	0.0155	0.0345	0.000002
<i>h</i>	-1 + Gamma(18.01, 12)	-1 + G(85%)	0.5	0.3535	70.69	0.85	0.0629	0.0871	0.0629	0.0867	0.0003
<i>i</i>	-1 + Gamma(11.445, 7.63)	-1 + G(75%)	0.5	0.4434	88.68	0.75	0.1184	0.1317	0.1184	0.1286	0.0030
<i>j</i>	Expo(1/0.5)	E(86%)	0.5	0.5	100	0.864	0	0.1353	0	0.1170	0.0183
<i>k</i>	-1 + Expo(1/1.5)	-1 + E(11%)	0.5	0.5	100	0.117	0.8647	0.0183	0.8647	0.0158	0.0025

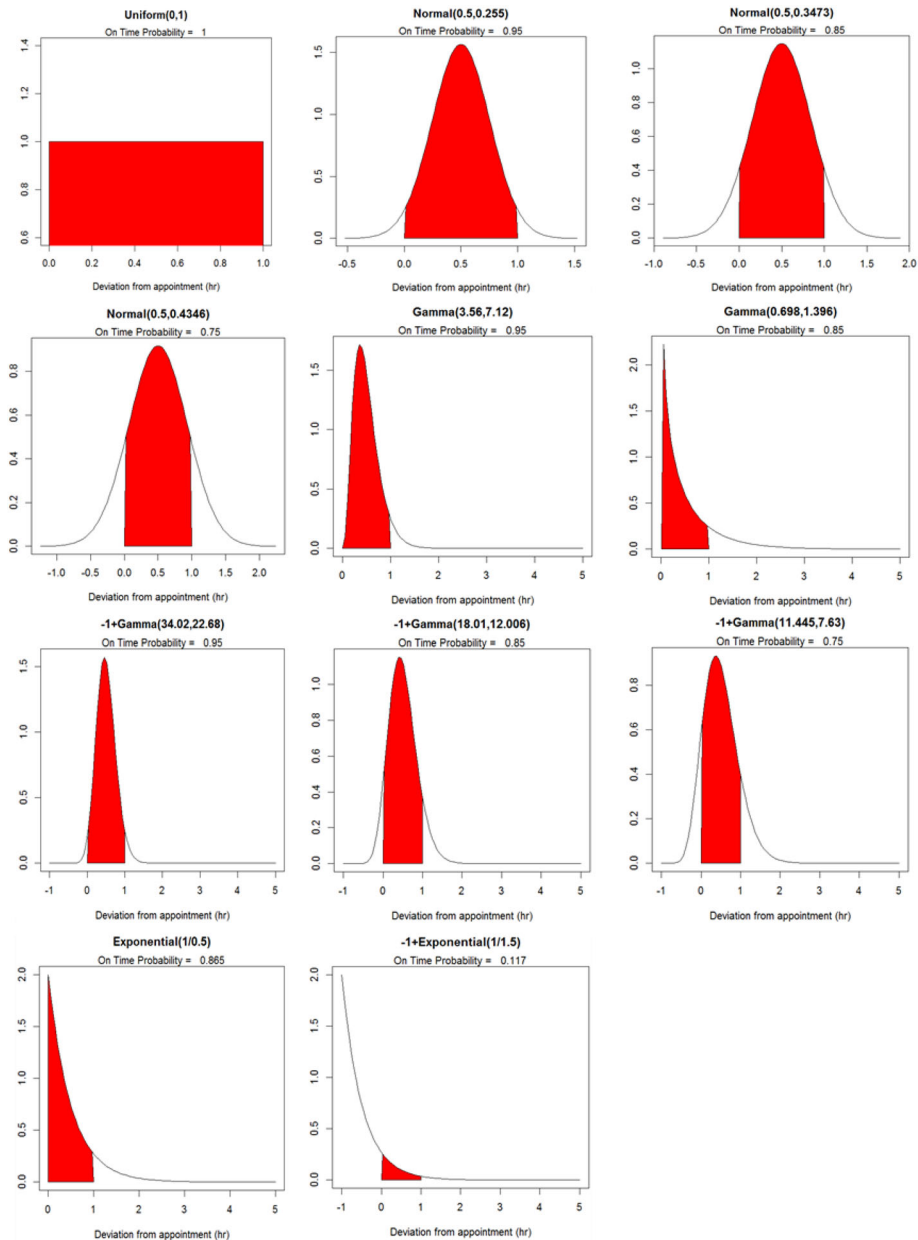


Fig. 6 Probability distribution functions of the deviation from appointment time

Figure 6 shows the probability distribution function of scenarios *a* to *k*. These scenarios analyze situations where a significant fraction of trucks do not comply with the appointment time considering that some ports where a TAS is implemented exhibit more than 97% of on time arrivals, such as the Port of Sydney, Australia (New South Wales Government 2014).

The baseline and the eleven scenarios described above for each configuration of TAS schedule were simulated for 50 replications. Each replication consists of 4 days of operation

Table 3 95% Confidence interval on the performance measures related to gate congestion of the baseline scenario (random arrivals and 8 cranes)

Performance measure	Estimated mean	Half width
Mean truck waiting time (min)	79.36	4.63
Maximum truck waiting time (min)	797.09	10.60
Mean truck queue length	9.80	0.59
Maximum truck queue length	102.94	2.44
Mean crane utilization (%)	54.18	0.41

receiving a total of 658 containers to be loaded onto 6 ships. For all the 95% confidence intervals computed for the performance measures related to gate congestion, this number of replications achieved an average relative precision of 6.21%. Furthermore, the relative precision of 98.22% of all the confidence intervals is smaller than 15%.

Table 3 shows the 95% confidence intervals of the estimated performance measures related to gate congestion and crane utilization of the baseline scenario.

Table 3 shows that, despite having a moderate mean crane utilization, the estimated mean truck waiting time exceeds 1 h and this result is similar to the one reported by TPA (see Sect. 3). Furthermore, in extreme cases, the waiting time can be over 13 h. These cases occur when trucks arrive near the end of the day and have to wait until the next day to be served by a crane. Similarly, the mean truck length is about 10 trucks, which is a reasonable size for TPA. However, in extreme cases like Day 2 (see Table 1), the maximum number of trucks in the queue exceeds 100.

In order to compare the performance of a TAS with the baseline, Fig. 7 shows the 95% confidence intervals for the mean and maximum truck waiting time for each scenario described in Table 2 and for both schedule policies, considering the same number of cranes. This figure shows a significant reduction on the waiting time using a TAS. Furthermore, the results show that an even number of appointments per day (TAS/Day-Even) outperforms the system where the number of appointments varies per day (TAS/Day-Var). For the schedule policy TAS/Day-Even, in the best case scenario, with full appointment compliance ($U(100\%)$), the mean waiting time is < 1 min and the maximum waiting time is approximately 20 min only. However, if truck arrivals deviate from the appointment time, the mean waiting time is still below 10 min for most cases, including the scenarios where 75% of the trucks comply with the appointment time. Furthermore, for the extreme case $-1 + E(11\%)$, the mean waiting time is below 15 min. For the schedule policy TAS/Day-Var, the mean waiting time is between 20 and 25 min for any deviation from the appointment time, which implies approximately 1-h reduction on the mean waiting time of the baseline. Regarding the maximum waiting time, the larger the appointment compliance, the smaller the maximum waiting time. For most of the scenarios of the deviation from the appointment time, the maximum waiting time is about 5 h smaller than the maximum time in the baseline scenario. However, the maximum waiting time could be only 20 min for the best case scenario ($U(100\%)$ with TAS/Day-Even).

The TAS has a similar impact on queue length. The mean and maximum truck queue lengths for the TAS/Day-Even across the eleven deviation scenarios are 0.4 and 6.4, respectively. On the other hand, the mean and maximum truck queue lengths for the TAS/Day-Var across the eleven deviation scenarios are 2.65 and 28.34, respectively. Furthermore, in the worst case scenario, the maximum truck queue length is around 30 trucks, which is a significant reduction on congestion on the gate and near highways.

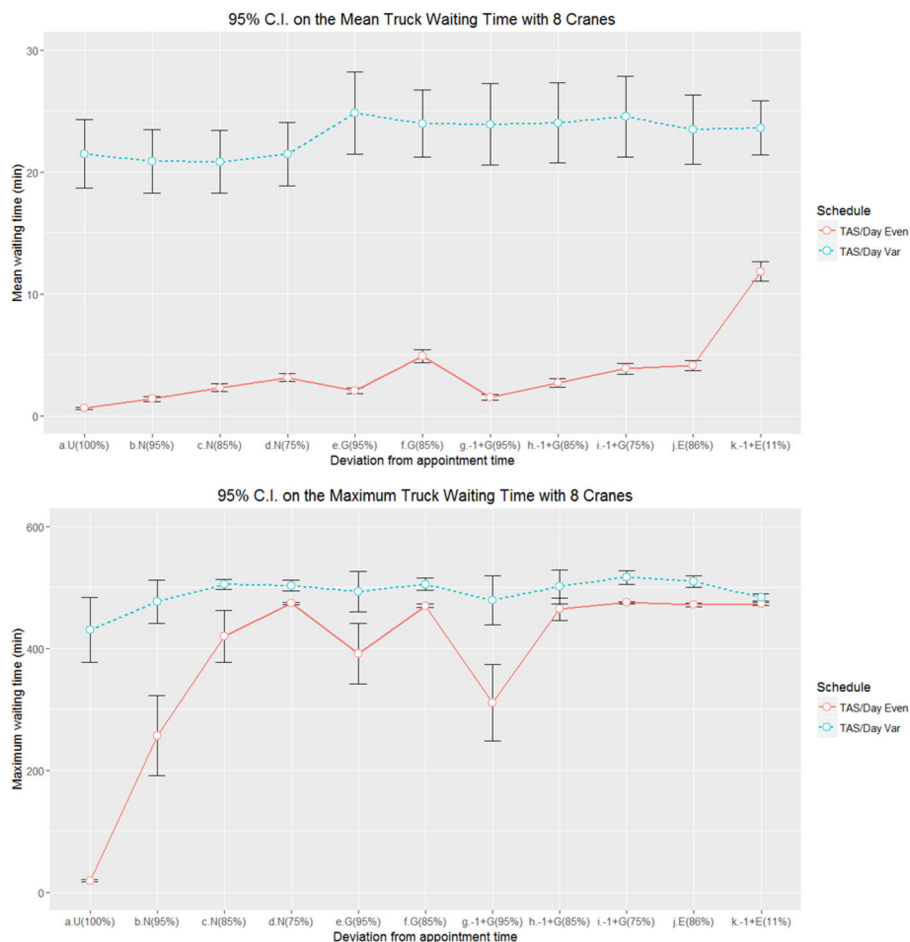


Fig. 7 95% CI on the mean (*above*) and maximum (*below*) truck waiting time with a TAS

Given 8 cranes, the average mean crane utilization for the TAS across the eleven deviation scenarios are 53.75% for TAS/Day-Even and 53.86% for TAS/Day-Var, which imply a non-significant reduction on crane utilization, compared with the baseline scenario.

Given the significant reduction on the waiting time using a TAS, the port might be able to reduce the number of cranes while keeping the gate congestion at a reasonable level. In order to analyze how many cranes can be taken out of the system while the waiting time is still reduced, the simulation model was executed decreasing the number of cranes in steps of one. The results suggest that the minimum number of cranes required to implement a TAS/Day-Even and still reduce the waiting time is 5 (−3 cranes); meanwhile the minimum number of cranes required by the TAS/Day-Var is 7 (−1 crane). If the number of cranes is further reduced, the waiting time increases above the baseline level. This result implies that TAS/Day-Even not only outperforms TAS/Day-Var in terms of gate congestion, but also allows the port operating with fewer resources without having a significant impact on the performance at the gate. This is particularly important at the port terminal because some cranes are outsourced. Consequently, reducing

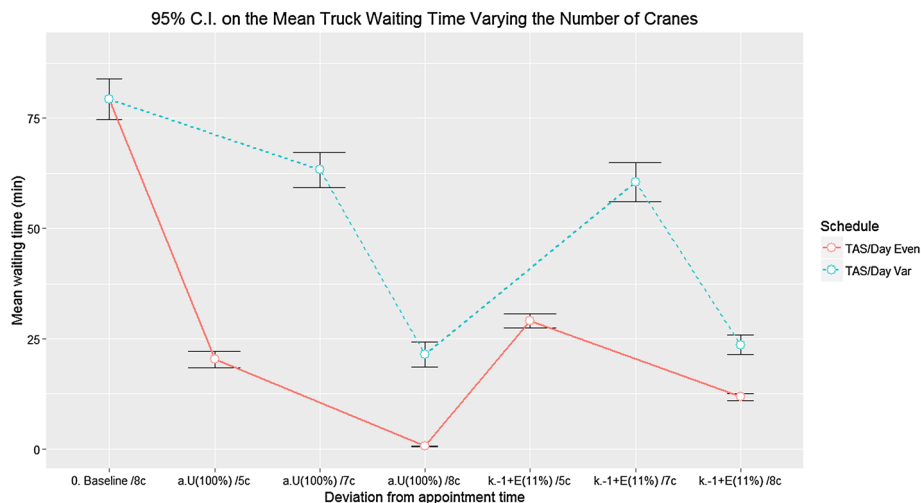


Fig. 8 95% CI on the mean truck waiting time for the baseline and best/worst TAS compliance scenarios varying the number of cranes

the number of cranes being used at the gate/yard would imply a reduction on operation cost.

Figure 8 shows the 95% confidence intervals on the mean truck waiting time for the baseline, the worst compliance TAS scenario ($-1+E(11\%)$) and the best compliance TAS scenario ($U(100\%)$) for both schedule policies and varying the number of cranes. This figure shows that still in the worst case scenario using TAS/Day-Var and reducing the number of cranes by one, the mean waiting time is reduced from 80 to 63 min. Furthermore, reducing the number of cranes by this amount keeps the crane utilization level around 61%. On the other hand, in the worst case with the TAS/Day-Even scenario, the number of cranes can be reduced up to 5 and still reduce the mean waiting time from 80 to 30 min, while keeping a crane utilization level around 86%, which is very reasonable.

These results depict the significant benefit of implementing a TAS. Given that the main objective of a TAS is to avoid congestion at the gate by smoothing the arrivals over a day and/or over a week, the cranes, which are the bottleneck resource, can efficiently handle the arriving trucks maintaining a high service level.

5.2 Impact of TAS on container rehandling

Out of the 658 containers received over the four days of the experiments, 191 containers are full dry containers with an assigned ship. This is a condition required to evaluate the yard operations based on the rehandles incurred when loading the containers on the ship. Empty containers are stacked together and they do not require rehandles. Containers with other special characteristics, such as reefer containers, are out of the analysis because of the special treatment required for their movement. Out of the 191 containers, 107 are 20 ft containers and 84 are 40 ft containers. In order to evaluate the effect of a TAS on container rehandles in a stressed out scenario, the arrival of the 191 containers were scheduled to arrive in small time frames during the simulation. For the TAS/Day-Even, the containers are scheduled to arrive consecutively at the beginning of the simulation, while for TAS/Day-Var, the containers are scheduled to arrive during the busiest day (Day 2) of the simulation.

The simulation models of the baseline and the TAS produce a sequence of arrivals of containers that will be placed in the yard based on the allocation policy described in Sect. 3. Then, the heuristic procedure counts the number of rehandles incurred when loading these containers on their assigned ships. A rehandle is a non-value added movement performed when an interfering container must be temporarily relocated to reach another container under the former. The loading sequence used in the heuristic is based on the actual loading sequence observed at TPA during the period of data collection.

The effect of a TAS on the number of rehandles depends on the assignment of trucks/containers to appointment times. The following assignment policies were tested to compare them with the baseline scenario.

- a. Loading Sequence (LS). The arrivals of containers to be loaded on the same ship are scheduled to occur consecutively in the reverse order of the loading sequence. Thus, given the location policy shown in Fig. 1, the first container loaded onto a ship is scheduled to arrive last, the second loaded container is scheduled to arrive before the last one, etc., until the last loaded container is scheduled to arrive first. When all containers of a given ship have been scheduled, then the containers of another ship are programmed. The analysis of the proximity of the appointment time with the loading time is not part of the scope of this paper. Therefore, for this paper we assume that, regardless the appointment time, all containers are scheduled to arrive before the loading operations start.
- b. Proportional Assignment (PA). For each appointment slot, the proportion of containers for different ships is the same as the overall proportion of containers. In addition, for the appointments given to containers of a specific ship, the LS is used.
- c. Random Sequence (RS). In this scenario, the real arrival sequence is used but smoothed using a TAS.

The RS assignment represents the case where the truck owner chooses its appointment time based on its preference. The LS represent the case where the port has complete information on the container characteristics and full control over the appointments. The PA could be seen as a mixture of the other two if some flexibility is granted to truck owners to pick from a small subset of appointment slots.

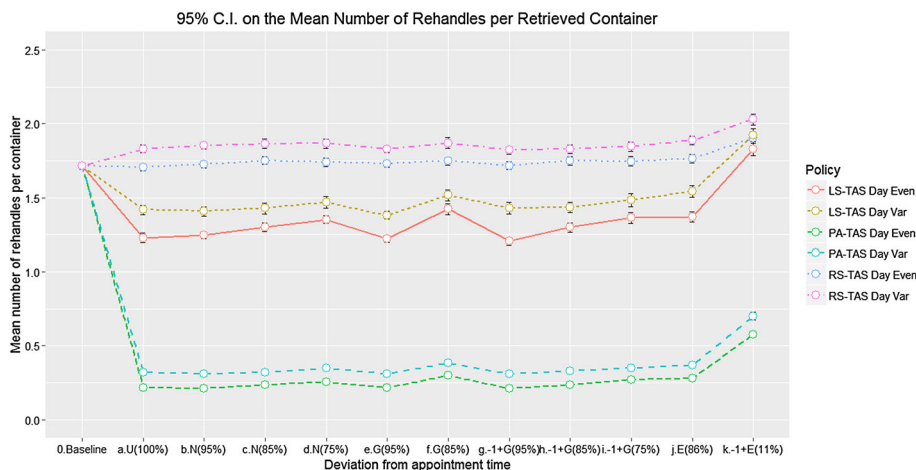
The following example explains the three types of assignment policies. Consider the following 10 pairs of container/ship: a/X, b/Z, c/Z, d/Y, e/Z, f/X, g/Z, h/Y, i/Z and j/Z. Assume that containers arrived in alphabetic order. In addition, assume that the loading sequence for ship X was: a, f; for ship Y was: h, d; and for ship Z was: i, c, b, e, j, g. Note that 60% of containers belong to ship Z and 20% for ships X and Y, each. Assume that there are three time slots (1, 2 and 3) in the TAS and the number of containers to be assigned to each slot is 3, 3 and 4, respectively. Table 4 shows the three policies of assignment of containers to time slots. Note that the overall proportion of containers to trucks is approximately the same within each time slot in the PA.

Note that using any of these policies does not impact the performance measures related to gate congestion, since they do not depend on the characteristic of the container that is carried by the truck. Nevertheless, the deviation from the appointment time and the TAS schedule policy impact the number of rehandles as well as the assignment policies. The heuristic procedure described in Sect. 4 was fed with each arrival sequence obtained from the discrete-event simulation model for each replication and for each scenario.

Figure 9 shows the 95% CI on the mean number of rehandles per retrieved container for the baseline and the different deviation scenarios and for each combination of TAS schedule policy and appointment assignment policy. Note that the mean number of rehandles in the baseline scenario is within the range of the estimation reported by the port terminal,

Table 4 Example of truck/container assignment policies in a TAS

Time slot	Programmed arrivals	Loading sequence (LS)	Proportional assignment (PA)	Random sequence (RS)
1	3	f/X, a/X, d/Y	f/X, g/Z, j/Z	a/X, b/Z, c/Z
2	3	h/Y, g/Z, j/Z	d/Y, e/Z, b/Z	d/Y, e/Z, f/X
3	4	e/Z, b/Z, c/Z, i/Z	a/X, h/Y, c/Z, i/Z	g/Z, h/Y, i/Z, j/Z

**Fig. 9** 95% CI on the mean number of rehables per retrieved container for each combination of TAS schedule policy and appointment assignment policy

as shown in Sect. 3. The following observations can be made from Fig. 9. First, this figure shows that, regardless the appointment assignment policy, the TAS/Day-Even schedule policy produces a smaller number of rehables than the TAS/Day-Var policy. Second, this figure shows that LS and PA policies significantly reduce the number of rehables, regardless the deviation from the appointment time and the TAS schedule policy. However, given a specific assignment policy, the number of rehables increases as the deviation from the appointment time increases. Third, if a TAS is implemented with RS, the number of rehables keeps at the same level than the baseline scenario and it could increase in an extreme scenario where trucks significantly deviate from the appointment time ($-1+E(11\%)$). The PA policy produces the most significant reduction on the number of rehables. Even in its worst case scenario, the number of rehables is almost three times smaller than the baseline scenario. On the other hand, we did not find a significant difference on the mean number of rehables reducing the number of cranes to the levels tested in Sect. 5.1

These results show that assigning appointments to containers of different ships within time slots reduces the negative effect of early and late arrivals. In general, the results suggest that implementing a TAS with an intelligent appointment assignment policy can not only reduce congestion at the gate, but also can significantly reduce non-value added movements when loading containers on ships.

6 Managerial insights

This section presents a discussion on the potential implementation issues of a TAS at the port terminal of TPA and the limitations of this paper.

The results presented in the previous section are limited to a relatively small sample data of 658 arrivals of containers and 191 containers loaded on ships. However, this sample includes several features of the real system, such as containers belonging to different ships, uneven arrival patterns across the hours of a day and across the days of the week. Regardless these limitations, the observations drawn from the results shown in Sect. 5 are systematic and the behavior would hold on a wide variety of scenarios.

In summary, the results shown in Sect. 5 suggest that a TAS accompanied of an adequate policy to assign appointments to trucks can significantly reduce congestion at the gate and rehandles at the yard. However, the following challenges may appear when implementing a TAS such as the one analyzed in this paper:

- Having accurate information of incoming containers. The port may know the ship and size of a container in advance to its arrival. But, there are other important attributes that would not be always known until the container is at the port. Therefore, it would be very useful for ports implementing distributed information systems to track the containers before their arrival.
- Definition of loading sequence in advance. A general loading sequence may be defined based on the destination of the container. However, the real loading sequence is usually defined in real-time based on the weight of the containers and the potential unbalance of the cargo at the ship. Therefore, having an approximation of the weight of incoming containers would help to overcome this challenge.
- Timing. Implementing a TAS with the characteristics analyzed in this paper requires a tight coordination among different agents: port, truck drivers, ships, container owners, trade agents, etc. Therefore, business process management is needed to successfully implement a TAS due to the times and coordination required for different tasks.
- Appointment compliance. As observed in the results, the TAS would work if the trucks comply with their appointment time. However, it is necessary to assign appointments that both, port and truck drivers agree with. Therefore, penalty policies would be helpful to be defined for deviations from the appointment time and day, as long as the port terminals also offer service standard levels in terms of truck turnaround times (such as in the Port of Sydney in Australia). In addition, to complement the implementation of a TAS it could be considered the use of a truck center or truck marshaling area such as those employed by the ports in Australia that may be used especially for those trucks arriving earlier to their appointment or even for the late arrivals that need to be re-assigned a time slot period to their service.
- Definition of TAS schedule policy. Even though the previous section shows that TAS/Day-Even outperforms TAS/Day-Var in all gate and yard performance measures, the latter may be more appropriate to implement since the port could also identify peak, shoulder and off-peak days, similar to the patterns identified and implemented over a day.

As observed in the challenges enumerated above, there are a lot of opportunities to use data analytic tools to design a TAS where all the different agents involved in the operations agree with and in order to improve the operations at the port and deliver a high service level to truck drivers, container owners and ships.

7 Conclusions and recommendations for further research

In this paper, we study the impacts of implementing a truck appointment system in a port terminal, considering the particular case of the Port of Arica. The analysis includes performance metrics for yard operations as well as congestion measures.

The results show a significant reduction on waiting time (mean and maximum) and queue length when using a TAS, even if the gate demand per day varies across the week. The reduction on truck waiting time implies shorter turnaround times for trucks and less congestion in the port area and highways around the port, which could increase overall safety for the drivers. Moreover, implementing a TAS could reduce operational cost by reducing the number of resources and still keep a satisfactory service level. However, the TAS requires an appointment assignment policy that is aligned with the loading sequence to achieve a significant reduction of non-value added movements of containers in the yard.

While a TAS may significantly improve the operations at a port, its design and implementation is not trivial. It requires combining multiple methods and techniques, such as business process management to align the tasks and times of the multiple agents that are involved in port operations. Furthermore, data analytics is also required to predict attributes of containers before they arrive.

The simulation model presented in this work provides useful information for the port terminal managers who have to take the decision whether to implement this type of system and who require having more insights of the implications and benefits. With the model proposed here, the managers are able to evaluate different scenarios and analyze the benefits prior to implementing any system or mechanism that affects several users of the port. Although we know that spreading demand can reduce delays, it is important to evaluate different scenarios and analyze the benefits prior to implementing any system or mechanism that affect several users of the Port. In addition, for the specific case of the Port of Arica, due to the political issues that exist between the two countries, port managers require this type of information to discuss it with the users of the port prior to implementing any type of system or mechanism that could affect or change current operations and specifically, truck arrivals. The port authority may be also interested in this information as a regulator that aims to have better operations at the port system and is also able to visualize bottlenecks and opportunities to improve the operations at the port.

For further research we propose to evaluate other configurations of truck appointment systems in which different operational rules may be analyzed in order to support gate capacity planning and rules related to the cases, for instance, in which a truck arrives late. Given that an appointment reduces the truck operators' flexibility, it is important to also measure the impacts on their productivity and utilization of their fleet in the current situation and with an appointment system. As a complement to the appointment system, a notification and booking system could also be simulated in order to evaluate under different scenarios and operational rules, in which situations each type of system provides better results in order to give recommendations for the port managers. Furthermore, a mechanism can be studied to enhance round trips at the port terminal to reduce empty truck trips and in this way reduce congestion and increase the productivity of the drayage operations. Finally, we recommend to study under which conditions, implementing a TAS may result in a bad policy for the port terminal and which other alternatives could be used to coordinate truck arrivals.

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